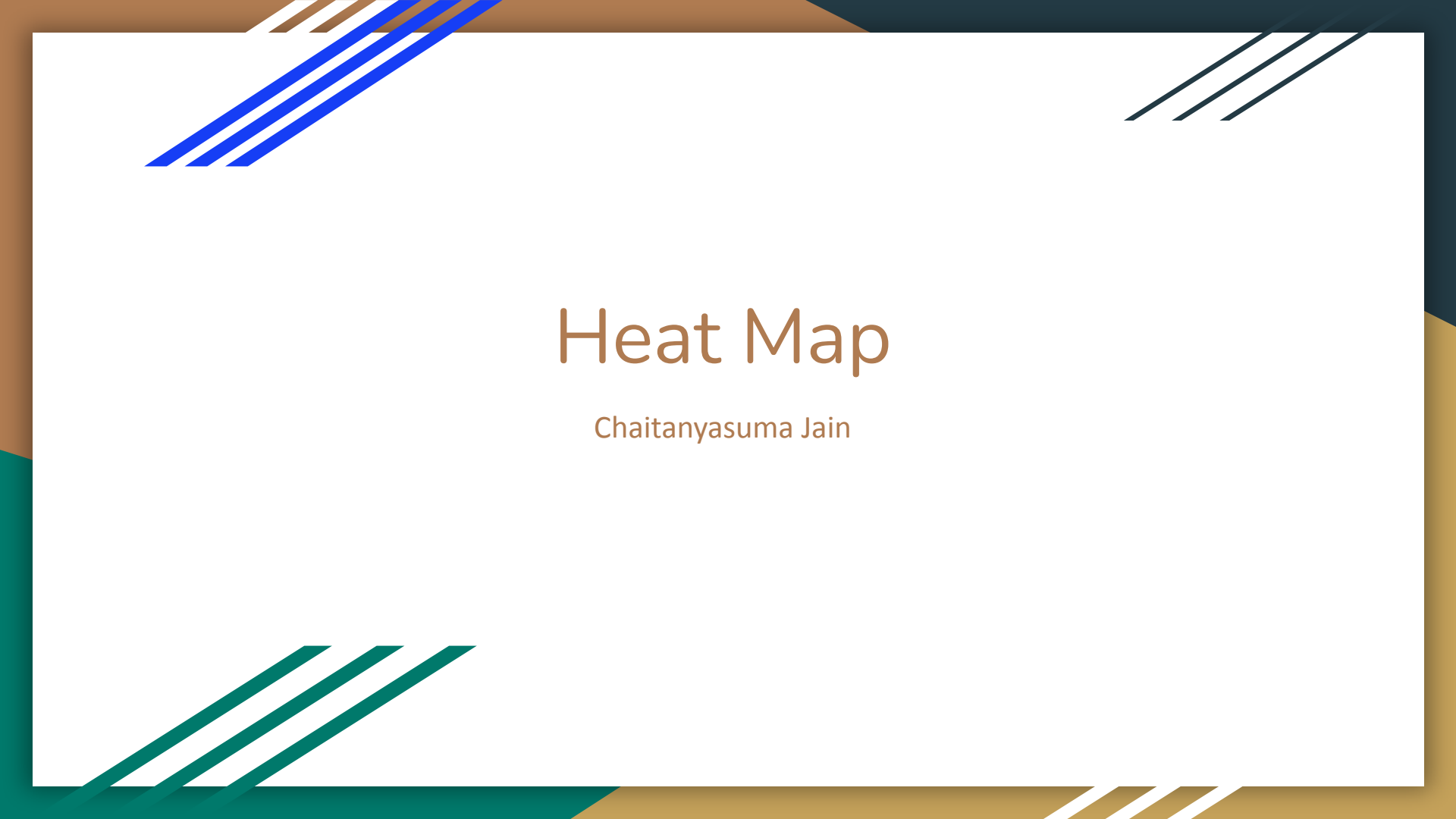


The slide features a white central area with decorative diagonal lines in blue, teal, and gold colors at the corners.

Heat Map

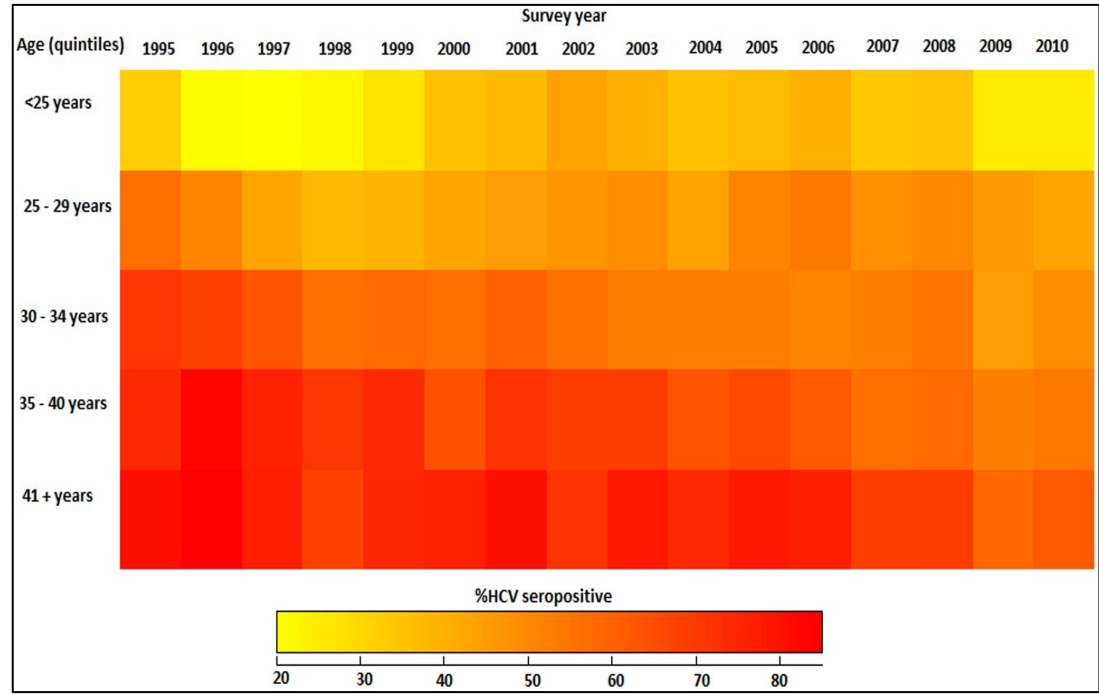
Chaitanyasuma Jain

Invention

- The concept of Heat Maps has existed since a long time with no particular information about who or what inspired it.
- But in 1991, Software designer **Cormac Kinney** trademarked the term 'heat map' to describe a 2D display depicting financial market information.

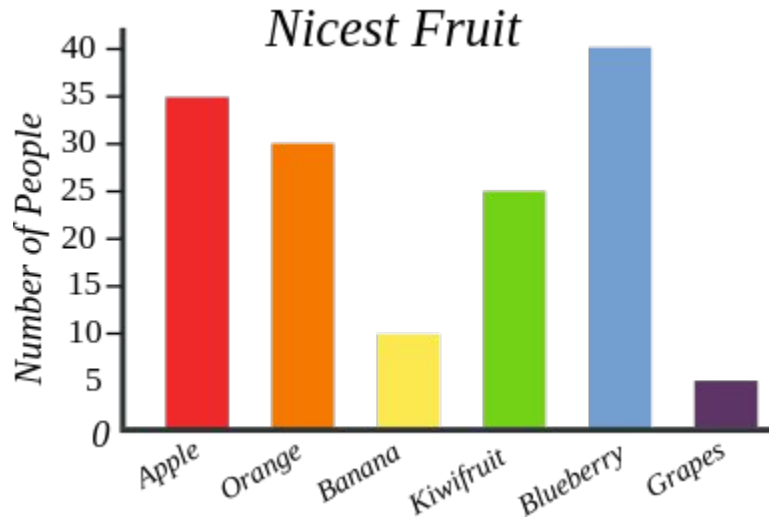
What is a Heatmap?

A heat map is a graphical representation of data where the individual values contained in a matrix are represented as colors (cell contouring)



Derived from

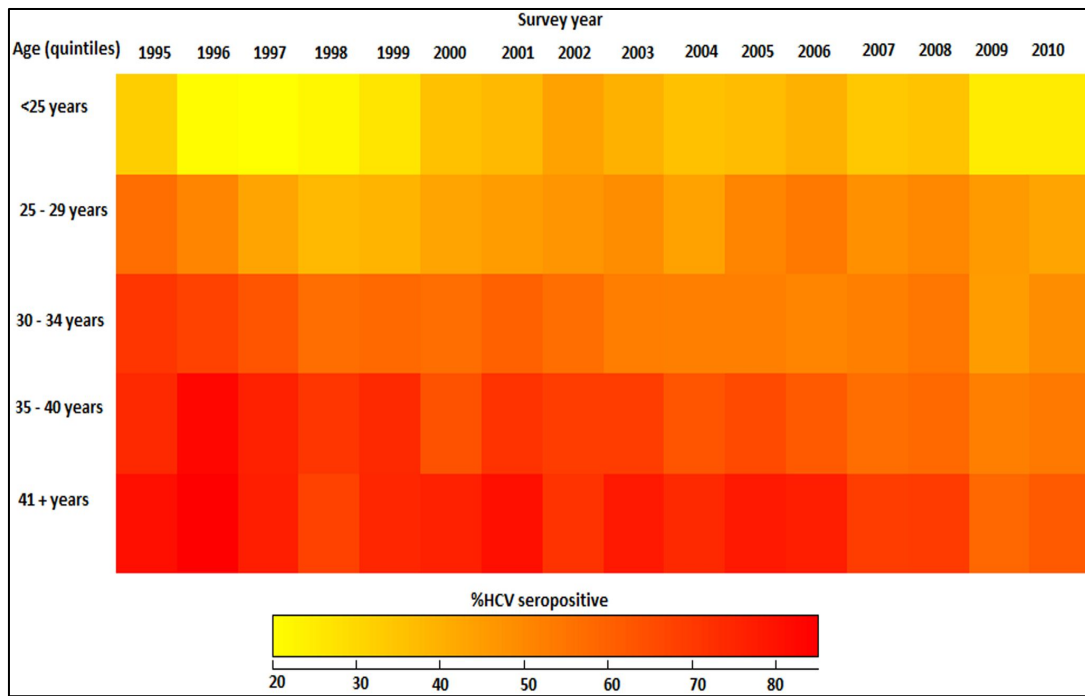
Bar Graph



How to read a Heat map?

Heatmaps are 2D graphs,
with 3 variables signified
by -

1. Row
2. Column
3. Color
4. Key



Real Life Applications

https://www.anychart.com/products/anychart/gallery/Heat_Map_Charts/,
<https://fancygrid.com/dashboards/heatmap-pollution/>, <https://datamatic.io/#id=0B3wq5VFm9PIlcU1qWHBZcUttVTg>,
<https://www.r-graph-gallery.com/heatmap/>

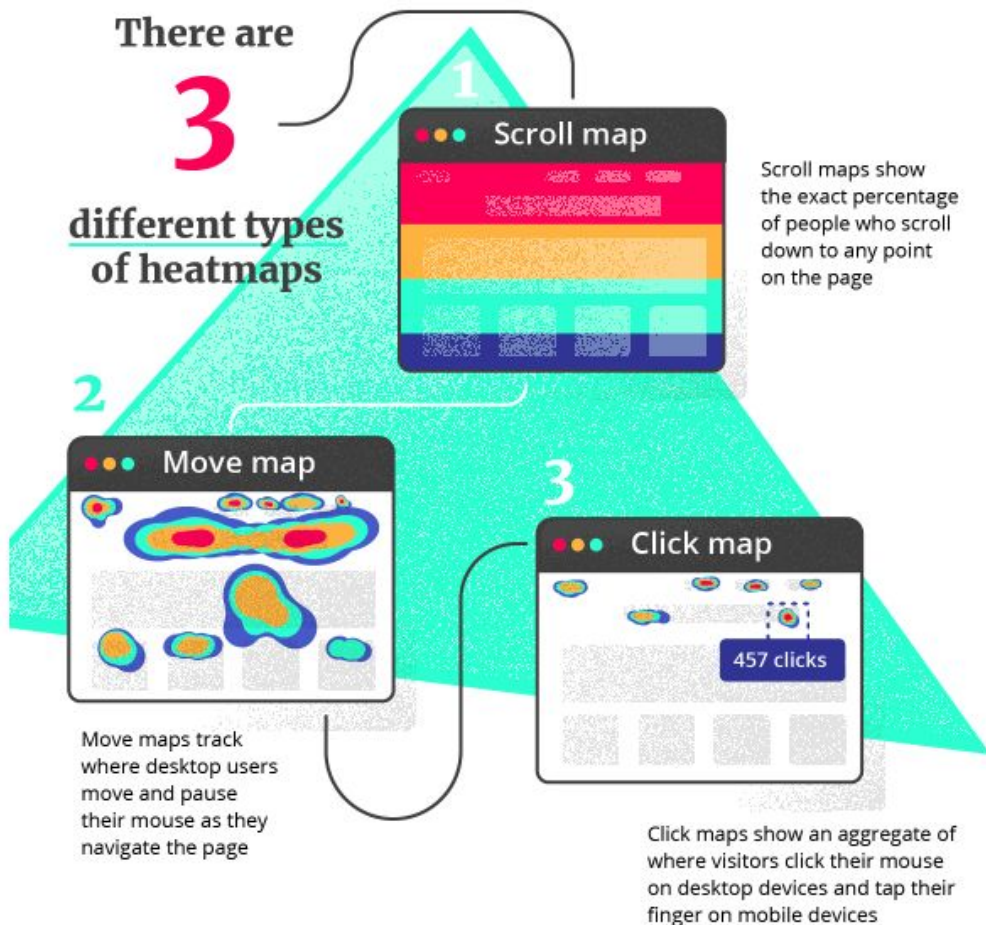
Also used for –

1. Website Redesign
2. UX and Usability Testing
3. Content Marketing

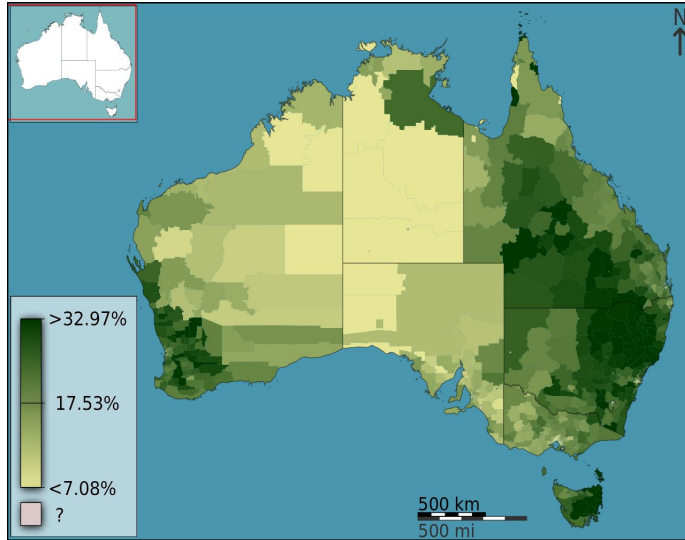
Implementation

- AnyChart : https://docs.anychart.com/Basic_Charts/Heat_Map_Chart
- AmChart : <https://www.amcharts.com/demos/heat-map-with-legend/>
- Gnuplot : <http://gnuplot.sourceforge.net/demo/heatmaps.html>,
<http://www.gnuplotting.org/heat-maps/>
- Tableau : https://onlinehelp.tableau.com/current/pro/desktop/en-us/buildexamples_density.htm
- Excel :
<https://www.accountingweb.com/technology/excel/how-to-heat-map-columns-of-numbers-in-excel>
- D3 : <https://observablehq.com/@randomfractals/deck-gl-heatmap>
- Fusion Chart (JSON) :
https://www.fusioncharts.com/dev/chart-guide/standard-charts/heat-map-chart#_layout
- Python : <https://python-graph-gallery.com/90-heatmaps-with-various-input-format/>

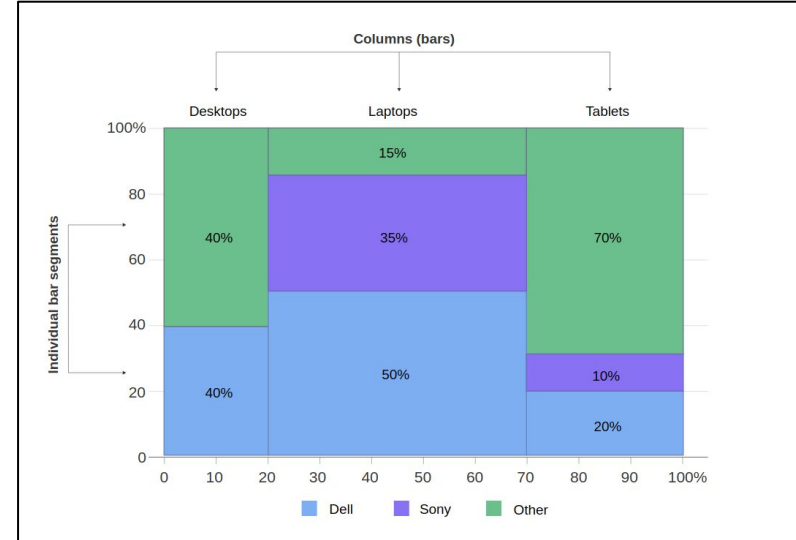
There are
3
different types
of heatmaps



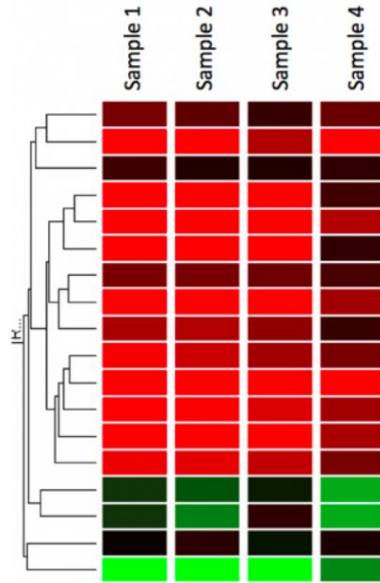
Variations & Similar Tools



A choropleth map that visualizes the fraction of Australians that identified as Anglican at the 2011 census



Marimekko Chart showing Electronics Sales



A biological heat map in which each row represents a gene, and red represents up-regulated genes and green represents down-regulated genes. Black represents unchanged expression.



Treemap

Functions

- Comparison without an axis
- Displaying data over time
- To display and find patterns
- To show relationships (biological heat map)

Advantages & Disadvantages

Advantages :

1. Heatmaps are a chart better suited to displaying a more generalised view of numerical data
2. Effective graphics show clear conclusions
3. Visually appealing

Disadvantages :

1. It's harder to accurately tell the differences between colour shades and to extract specific data points from a heatmap
2. Cannot be used to replace tables

3D Modelling & Gamification

Colorful Puzzle Pieces can be used



3D Heat map

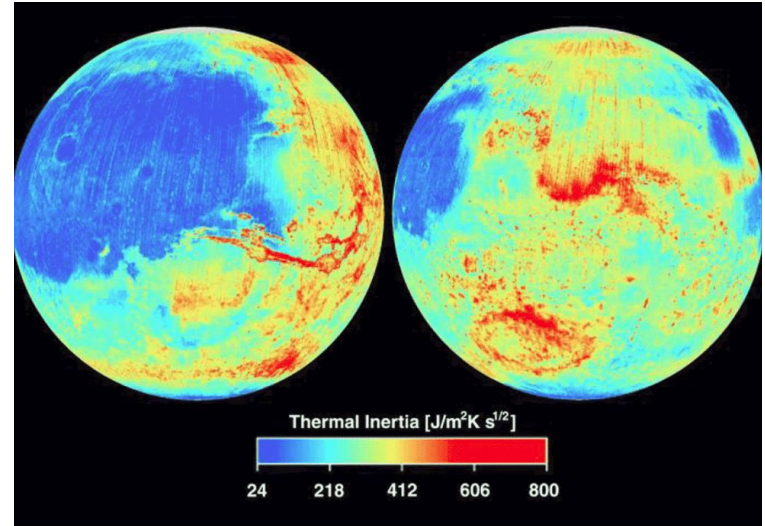
Data is plotted on a Heat Map chart in terms of latitude, longitude, and altitude. The data is plotted as a surface, with a physical dimension as well as color representing altitude.

Examples :

<https://www.infragistics.com/help/winforms/chart-working-with-3d-heat-map-chart-data>

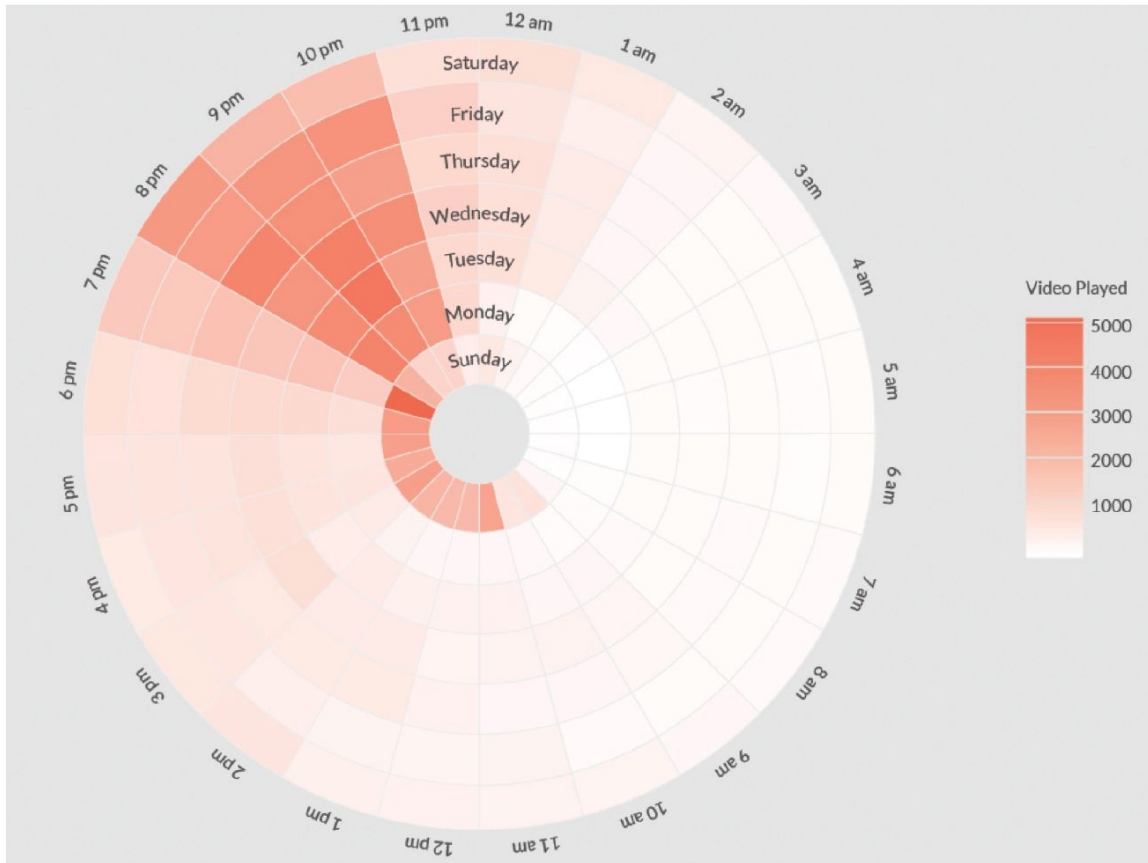
Implementation :

<https://stackoverflow.com/questions/40853556/3d-discrete-heatmap-in-matplotlib>



Circular Heat Map

A radial heat map is a variation of heat map, where the task is aligned radially.



Need & Advantage

With radial mapping, you're not only able to visualize your data on a map but you can also easily and immediately understand where the majority of your activity is located (map overlays)

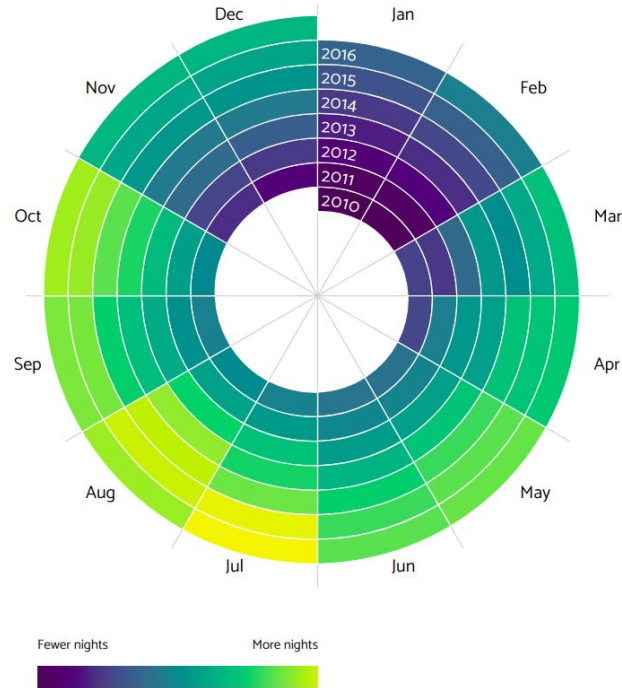
Implementation

- R : <https://rdr.io/cran/circular/man/heatmap.circular.html>
- HTML CSS : <http://blocks.org/arpitnarechania/bobdc4f9a377ea9d8612677e12f65b82>
- GNUPlot : <http://www.gnuplotting.org/circular-heat-map/>
- Python :
<https://stackoverflow.com/questions/25236026/circular-interpolated-heat-map-plot-using-python>
- MATLAB :
<https://in.mathworks.com/matlabcentral/answers/49287-how-to-create-a-circular-heat-map>

Spiral Heat Map

A spiral heat map is a specific type of heat map designed for continuous comparable cycles like years or days. The spiral design makes it possible to compare the cycles, but keeping the continuous timeline along the spiral.

An example of a spiral heatmap, showing the number of tourist nights in Berlin (from the Makeover Monday series).



Technical Papers

https://en.wikipedia.org/wiki/Heat_map#Types